8 **(a)** Fig. 8.1 shows a circuit.

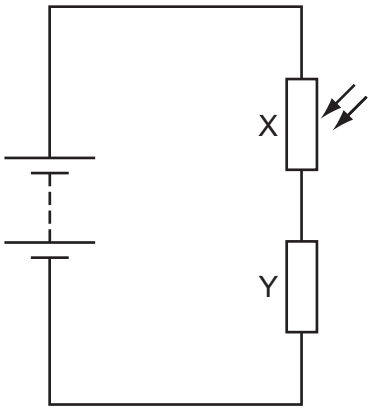


Fig. 8.1

(i) State the name of component X.

..... [1]

(ii) The potential difference (p.d.) across component Y is measured with a voltmeter.

On Fig. 8.1, draw the symbol for the voltmeter and its connections to the circuit. [1]

(iii) The electromotive force (e.m.f.) of the battery is 12 V.

Component Y has a resistance of 400 Ω.

In a brightly lit room, the resistance of component X is 350 Ω.

1. Calculate the current in the circuit.

current = ... [2]

2. Calculate the p.d. across component Y.

p.d. = ... [1]

(iv) In a dark room, the resistance of component X is very large.

State the effect this will have on the p.d. across component Y.

..... [1]

(b) Suggest a practical use for component X.

..... [1]

[Total: 7]

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